复分析第 1 次作业解答

Complex Analysis Homework 1: Solutions

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1. 复线性与 Cauchy–Riemann 方程 / Complex linearity and the Cauchy–Riemann equations

题目 / Problem. 设 $f = u + iv : \mathbb{C} \to \mathbb{C}$，$F = (u, v) : \mathbb{R}^2 \to \mathbb{R}^2$。证明若 f 全纯，则 J_F 是复线性映射，且 Cauchy–Riemann 方程等价于

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Let $f = u + iv : \mathbb{C} \to \mathbb{C}$ and $F = (u, v) : \mathbb{R}^2 \to \mathbb{R}^2$. Show that holomorphicity of f makes J_F complex-linear, and that the Cauchy–Riemann equations are equivalent to the displayed matrix identity.

中文解答. 令

$$A = J_F = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

在 $\mathbb{R}^2 \cong \mathbb{C}$ 下，J 正是“乘以 i”。实线性映射 A 为复线性，当且仅当 $A(iw) = iA(w)$ 对每个 w 成立，亦即 $AJ = JA$。直接相乘得到

$$JA = \begin{pmatrix} -v_x & -v_y \\ u_x & u_y \end{pmatrix}, \qquad AJ = \begin{pmatrix} u_y & -u_x \\ v_y & -v_x \end{pmatrix}.$$

两矩阵相等恰好等价于 $u_x = v_y$ 和 $u_y = -v_x$。若 f 全纯，则实微分为 $h \mapsto f'(z)h$，所以 J_F 复线性，以上等式也证明了所要求的矩阵表述。

English solution. Under $\mathbb{R}^2 \cong \mathbb{C}$, the matrix J represents multiplication by i. A real-linear map $A = J_F$ is complex-linear exactly when $A(iw) = iA(w)$ for every w, or equivalently $AJ = JA$. The displayed products show that this commutation relation is precisely $u_x = v_y$ and $u_y = -v_x$. If f is holomorphic, its real differential is $h \mapsto f'(z)h$, which is complex-linear. Thus its Jacobian commutes with J, and the matrix identity is equivalent to the Cauchy–Riemann equations.

2. 极坐标形式 / Polar-coordinate form

题目 / Problem. 证明 $f = u+iv$ 的极坐标 Cauchy–Riemann 方程为 $u_r = r^{-1}v_\theta$、$v_r = -r^{-1}u_\theta$。Show that the Cauchy–Riemann equations in polar coordinates take the form $u_r = r^{-1}v_\theta$ and $v_r = -r^{-1}u_\theta$.

中文解答. 以下取 $r>0$。由 $x=r\cos\theta$、$y=r\sin\theta$ 及链式法则，

$$u_r = u_x\cos\theta + u_y\sin\theta, \qquad \frac{v_\theta}{r} = -v_x\sin\theta + v_y\cos\theta.$$

代入 $u_x=v_y$、$u_y=-v_x$，便得 $u_r = r^{-1}v_\theta$。同理，

$$v_r = v_x\cos\theta + v_y\sin\theta, \qquad -\frac{u_\theta}{r} = u_x\sin\theta - u_y\cos\theta,$$

故 $v_r = -r^{-1}u_\theta$。由于从 (r,θ) 到 (x,y) 的雅可比行列式为 $r\neq 0$，这些极坐标方程也反过来蕴含直角坐标的 Cauchy–Riemann 方程。

English solution. For $r>0$, the chain rule gives the four identities displayed above. Substituting $u_x=v_y$ and $u_y=-v_x$ yields

$$\boxed{u_r = \frac{1}{r}v_\theta, \qquad v_r = -\frac{1}{r}u_\theta.}$$

Conversely, the polar change of variables has Jacobian determinant $r\neq 0$; reversing the same calculation recovers the Cartesian Cauchy–Riemann equations. The formulas are not asserted at $r=0$, where polar coordinates are singular.

3. Laplace 算子与 Wirtinger 导数 / Laplacian and Wirtinger derivatives

题目 / Problem. 证明 $\Delta = 4\partial_z\partial_{\bar z}$。Show that $\Delta = 4\partial_z\partial_{\bar z}$.

中文解答. 对任意 C^2 函数，定义

$$\partial_z = \frac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\bar z} = \frac{1}{2}(\partial_x + i\partial_y).$$

由于混合偏导可交换，

$$4\partial_z\partial_{\bar z} = (\partial_x - i\partial_y)(\partial_x + i\partial_y) = \partial_x^2 + \partial_y^2 = \Delta.$$

English solution. For a C^2 function, expand the product of the two Wirtinger operators as above. The mixed terms cancel because $\partial_x\partial_y = \partial_y\partial_x$, leaving $4\partial_z\partial_{\bar z} = \Delta$.

4. 分式线性变换的分类 / Classification of fractional linear transformations

题目 / Problem. 设标准化的分式线性变换 $T(z) = (az+b)/(cz+d)$ 有有限不动点 z_1，其乘子为 $\kappa = T'(z_1)$。按题意，抛物型指 $\kappa=1$，双曲型指 $\kappa>0$ 且 $\kappa\neq 1$，椭圆型指 $|\kappa|=1$ 且 $\kappa\neq 1$，其余为斜向型。证明这四种情形依次等价于 $a+d=\pm 2$、$a+d\in\mathbb{R}$ 且 $|a+d|>2$、$a+d\in\mathbb{R}$ 且 $|a+d|<2$、以及 $a+d\notin\mathbb{R}$。题目提示还要求讨论不动点数、第二个不动点的乘子及 $\kappa+\kappa^{-1} = (a+d)^2 - 2$。

Let $T(z) = (az + b)/(cz + d)$ be normalized and have a finite fixed point z_1. Its multiplier is $\kappa = T'(z_1)$. The stated definitions are: parabolic for $\kappa = 1$; hyperbolic for positive real $\kappa \neq 1$; elliptic for $|\kappa| = 1$, $\kappa \neq 1$; and loxodromic otherwise. Prove that these correspond, respectively, to trace $a + d = \pm 2$, real trace with $|a + d| > 2$, real trace with $|a + d| < 2$, and nonreal trace. The hints concern the number of fixed points, the multiplier at a second fixed point, and $\kappa + \kappa^{-1} = (a + d)^2 - 2$.

中文解答. “标准化”指 $ad - bc = 1$。设 $t = a + d$、$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{C})$。若有限不动点 z_1 的对应特征值是 $\lambda = cz_1 + d$，则

$$M\begin{pmatrix} z_1 \\ 1 \end{pmatrix} = \lambda \begin{pmatrix} z_1 \\ 1 \end{pmatrix}, \qquad \kappa = T'(z_1) = \frac{ad - bc}{(cz_1 + d)^2} = \lambda^{-2}.$$

另一特征值是 λ^{-1}，因此

$$\boxed{\kappa + \kappa^{-1} = \lambda^{-2} + \lambda^2 = t^2 - 2.} \tag{1}$$

若有两个不同不动点，它们的乘子互为倒数；第二个点为 ∞ 时，在局部坐标 $w = 1/z$ 中计算其乘子即可。若 $\kappa = 1$，则 $t^2 = 4$，矩阵有重特征值；除恒等变换外，T 恰有一个不动点。

由式 (1) 可逐一判定：

$$\begin{aligned} \kappa = 1 &\iff t = \pm 2; \\ \kappa > 0,\ \kappa \neq 1 &\iff t \in \mathbb{R},\ |t| > 2; \\ |\kappa| = 1,\ \kappa \neq 1 &\iff t \in \mathbb{R},\ |t| < 2. \end{aligned}$$

例如第二行中 $\kappa + \kappa^{-1} > 2$，故 $t^2 > 4$；反向则由 $X + X^{-1} = t^2 - 2 > 2$ 的两个根均为正实数可得。第三行对应 $\kappa + \kappa^{-1} \in [-2, 2)$，故 $t^2 \in [0, 4)$；反向则两根在单位圆上。四类穷尽所有乘子，所以剩余的斜向型情形恰为 $t \notin \mathbb{R}$。

English solution. Write $t = \operatorname{tr} M$ and normalize M by $\det M = 1$. At a finite fixed point, $(z_1, 1)^T$ is an eigenvector with eigenvalue $\lambda = cz_1 + d$, and $\kappa = T'(z_1) = \lambda^{-2}$. The other eigenvalue is λ^{-1}, so equation (1) follows. Two distinct fixed points therefore have reciprocal multipliers (use $w = 1/z$ when one is at infinity). The alternatives now follow from the ranges of $\kappa + \kappa^{-1}$: it equals 2 for $\kappa = 1$, lies in $(2, \infty)$ for positive real $\kappa \neq 1$, and lies in $[-2, 2)$ for unit-modulus $\kappa \neq 1$. Conversely, the quadratic equation $X + X^{-1} = t^2 - 2$ gives the indicated type in each real-trace range. All remaining cases have nonreal trace and are loxodromic.

题目提示的例外 / Exception in the hint. 恒等变换也满足 $\kappa = 1$ 和 $t = \pm 2$，但它的每个点都是不动点。因此“$\kappa = 1$ 当且仅当只有一个不动点”须排除恒等变换；提示中第二个不动点处的 $f'(z_2)$ 应为 $T'(z_2)$。The identity has multiplier 1 and trace ± 2 but fixes every point; the unique-fixed-point claim needs this exception. The derivative in the second hint should be $T'(z_2)$.

5. Schwarzian 导数 / Schwarzian derivative

题目 / Problem. 对全纯函数 f，其 Schwarzian 导数为

$$S(f) = \frac{f'''}{f'} - \frac{3}{2}\left(\frac{f''}{f'}\right)^2.$$

证明：(1) $S(f) \equiv 0$ 当且仅当 f 是分式线性变换；(2) 对全纯 g，有 $S(g\circ f) = (S(g)\circ f)(f')^2 + S(f)$，特别地，若 g 是分式线性变换，则 $S(g \circ f) = S(f)$。提示：令 $N(f) = f''/f'$，则 $S(f) = N(f)' - \frac{1}{2}N(f)^2$。

For a holomorphic f, define $S(f) = f'''/f' - \frac{3}{2}(f''/f')^2$. Show that (1) $S(f) \equiv 0$ exactly for fractional linear maps and (2) $S(g \circ f) = (S(g) \circ f)(f')^2 + S(f)$ for holomorphic g; in particular, postcomposition by a fractional linear map leaves $S(f)$ unchanged. Hint: $N(f) = f''/f'$ satisfies $S(f) = N(f)' - \frac{1}{2}N(f)^2$.

(1) $S(f) \equiv 0$ 与分式线性变换 / Vanishing Schwarzian and Möbius maps

中文解答. 在连通定义域上先假设 $f' \neq 0$，令 $N = f''/f'$。直接求导得

$$S(f) = N' - \frac{1}{2}N^2.$$

若 $S(f) = 0$，则 $N' = N^2/2$。如果 N 在某点为零，微分方程解的唯一性说明 $N \equiv 0$，于是 $f(z) = Az + B$，其中 $A \neq 0$。否则 N 处处非零，且

$$\left(\frac{1}{N}\right)' = -\frac{1}{2}, \qquad N(z) = -\frac{2}{z-\zeta}$$

（常数 ζ 不在定义域内）。积分两次得到 $f'(z) = C(z-\zeta)^{-2}$ 与 $f(z) = A(z-\zeta)^{-1} + B$，也是分式线性变换。反过来，对于 $f(z) = (az+b)/(cz+d)$，

$$f'(z) = \frac{ad-bc}{(cz+d)^2}, \qquad N(f) = -\frac{2c}{cz+d}, \qquad N(f)' = \frac{2c^2}{(cz+d)^2} = \frac{1}{2}N(f)^2,$$

故 $S(f) = 0$。

English solution. On a connected domain where $f' \neq 0$, put $N = f''/f'$. Then $S(f) = N' - N^2/2$. If $S(f) = 0$, the differential equation $N' = N^2/2$ has either the zero solution, giving the affine map $Az+B$, or a nowhere-zero solution $N = -2/(z-\zeta)$, giving $f = A/(z-\zeta)+B$. Both are fractional linear maps. Conversely, the displayed derivative calculation for $f = (az+b)/(cz+d)$ gives $N' = N^2/2$, hence $S(f) = 0$. If critical points of f were initially allowed, a zero Schwarzian on the regular set actually rules them out: at a zero of f' of order $m \geq 1$, $S(f)$ has leading term $-m(m+2)/(2(z-z_0)^2)$.

(2) 复合公式 / Composition law

中文解答. 在相关导数非零处，设 $A = (N(g)\circ f)f'$、$B = N(f)$。链式法则给出 $N(g\circ f) = A+B$，且

$$\begin{aligned} S(g\circ f) &= A' + B' - \tfrac{1}{2}(A+B)^2 \\ &= \big(N(g)' - \tfrac{1}{2}N(g)^2\big)\circ f\,(f')^2 + \big(B' - \tfrac{1}{2}B^2\big) + (N(g)\circ f)\big(f'' - f'B\big) \\ &= (S(g)\circ f)(f')^2 + S(f), \end{aligned}$$

因为 $f'B = f''$。若 g 是分式线性变换，由第 (1) 问 $S(g) = 0$，遂有 $S(g\circ f) = S(f)$。

English solution. The chain rule gives $N(g\circ f) = (N(g)\circ f)f' + N(f)$. Expanding $N(g\circ f)' - \frac{1}{2}N(g\circ f)^2$ as above, the cross term cancels because $f'N(f) = f''$. Thus

$$\boxed{S(g\circ f) = (S(g)\circ f)(f')^2 + S(f).}$$

For a fractional linear g, part (1) gives $S(g) = 0$, so $S(g\circ f) = S(f)$.